Catherine McCaffrey

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Commonwealth Honors College
College of Natural Sciences
University of Massachusetts Amherst, Amherst MA

Education

- **University of Massachusetts Amherst, Amherst MA** — *Senior*
  BS in Astronomy, BS in Physics, BA in Science, Sustainability, and Cultural Inquiry — GPA: 3.7
- **University of Heidelberg, Heidelberg Germany** — *March 1st 2026-July 25th 2026*
  Graduate-Level (M.Sc.) Astronomy Coursework

Scientific Projects

- **Photometric Analysis of Local Starburst Galaxies: Dust and Star Formation Properties**
  Examined how differences in SFE between galactic centers and global rates results from galaxy morphology and environment
  - Tools & technologies used: Python, Mid/far-IR Spitzer, Herschel images
- **Identification and Characterization of Compact Red Galaxies ("Little Red Dots") at z $\approx$ 2**
  Developed a selection/modeling framework to classify LRD candidates at low–intermediate redshift near the Balmer break regime
  - Tools & technologies used: Python, Euclid spectroscopy and photometry (Euclid VIS, NISP + EXT)
- **Constraining Redshifts of DSFGs (in Abell 2744) with Medium Bands from UNCOVER**
  Using H-alpha detections in high-redshift DSFGs with medium bands to more effectively determine redshift
  - Tools & technologies used: Python, JWST UNCOVER/ ALMA spectroscopy
- **Improving Masses and Star Formation Rates in the Nearby Starburst NGC3351**
  Model FUV-IR/mm SED and constrain the total infrared emission and dust mass for this region to update the total SFR and Mass
  - Tools & technologies used: Python, Mid/far-IR Spitzer, Herschel and LMT/AzTEC images
- **Multi-Mode Quantum Memories for High-Throughput Satellite Entanglement Distribution**
  Proposing a more efficient satellite-based entanglement protocol to advance ultra-secure communication and quantum metrology
- **JWST PRIMER Reveals Increasing Specific Star Formation Rate across Cosmic Time**
  Analyzed approximately 54,000 galaxies to refine the evolution of star formation efficiency out to redshift 5
  - Tools & technologies used: Python, JWST PRIMER data
- **Observation and Color Magnitude Diagram of NGC 7789**
  Developed an image reduction and photometric analysis pipeline for the open cluster NGC 7789 to estimate age and distance.
  - Tools & technologies used: Python, Schmidt–Cassegrain telescope at the Amherst College Observatory (ACO)

Teaching and Research Experience

- **Five College Astronomy Department Summer Internship Program** — *June 1st 2026-August 10th 2026*
  Intern — Amherst, MA
  * Preparing a first-author manuscript analyzing how disparities in star formation efficiency (SFE) between galactic centers and global rates arise from galaxy morphology and environmental factors
- **Max Planck Institute for Astronomy** — *March 30th 2026-July 25th 2026*
  Intern — Heidelberg, Germany
  * Conducting research to identify "little red dot" candidates at low to intermediate redshift through joint fitting of Euclid NISP spectroscopy and photometry using spectral and photometric templates
  * Developing a robust selection and modeling framework to systematically identify and characterize these sources
- **Astronomy Outreach Instructor** — *February 2026*
  Volunteer — Sloatsburg, NY
  * Taught elementary and high school students core concepts in stellar evolution and galaxy formation
  * Designed interactive lessons and activities to promote engagement with astronomy and STEM
  * Encouraged scientific curiosity through hands-on learning and discussion
- **Teaching Assistance for Physics 132** — *Fall Semester 2025*
  TA — UMass Amherst
  * Assisted in weekly discussion and class sessions to teach key concepts in electricity, magnetism, radiation, optics, quantum theory, atomic structure, nuclear physics.
  * Held regular office hours, providing one-on-one guidance to students on coursework, homework, and exam preparation in a class of over 200 student
- **Lab Lead for Springfield Public School District** — *July 1st 2025-July 22nd 2025*
  Lab Lead — Springfield, MA

- Taught local school district teachers in grades K–12 about applicable astronomy activities for their classrooms
- Instructed teachers how astronomers extract information, like the age of a galaxy, using physical properties of stars, such as their color, and how they can use this information to create activities to bring back to their students

–**Five College Astronomy Department Summer Internship Program** *June 3rd 2024-August 8th 2024*
Intern Amherst, MA
- Improving Masses and Star Formation Rates in the Nearby Starburst NGC3351
- Collected mid/far IR+mm data, produced photometry/corrected photometry for PSF affects, as well as ran different models to my data

Professional Development Training

–**5-College Women+ Career Day** *January 31st 2026*
Presenter Smith College
- Presented ongoing research on star formation efficiencies in local star-forming galaxies, exploring how morphology and environment influence central vs. outer-arm star formation.

–**APS New England Section (NES) Annual Meeting** *November 7th 2025*
Presenter Brown University
- Presented ongoing research on star formation efficiencies in local star-forming galaxies, exploring how morphology and environment influence central vs. outer-arm star formation.

–**International Astronautical Congress** *September 29-October 3 2025*
Accepted Presenter Sydney, Australia
- Research on efficient satellite-based entanglement protocol to advance ultra-secure communication and quantum metrology accepted for presentation

–**Lee-Sip Research Conference** *August 9 2025*
Presenter University of Massachusetts Amherst
- Presented ongoing research on star formation efficiencies in local star-forming galaxies, exploring how morphology and environment influence central vs. outer-arm star formation.

–**Massachusetts Undergraduate Research Conference (MassURC)** *April 18, 2025*
Presenter University of Massachusetts Amherst
- Presented findings on galaxy mass and star formation in a peer-reviewed conference.

–**NASA Proposal Writing and Evaluation Experience (NPWEE) Academy** *Spring 2025*
Chief Scientist —
- Co-authored a proposal advocating for a Digital Twin System to monitor orbital debris.
- Reviewed and evaluated external proposals according to NASA guidelines.

–**APS Conferences for Undergraduate Women and Gender Minorities in Physics (CU*iP)** *January 24-26, 2025*
Presenter NYU
- Presented ongoing research involving SFR and mass update of Galaxy NGC3351, attending plenary talks and networking with invited speakers, facilitators, and peers.

Technical Skills and Interests

Programming Languages: Python, Latex
Computer software/ frameworks: QuickBooks, Excel, Microsoft 365
Coursework: Galactic and Extragalactic Astronomy, Physics and Chemistry of the ISM, Stellar Astronomy and Astrophysics, Classical Mechanics, Quantum Mechanics, Observational Techniques, Computational Physics
Areas of Interest: Link between gas, dust and star formation, link between individual stars and star clusters and the large star-forming structures of galaxies

Extra Cirriculars

–**Member** Euclid Consortium *Spring 2026*
–**Founder** UMass Amherst Campus Shine Initiative *Fall 2025-present*
–**Co-President** Amherst Quantum Initiative *Fall 2025-present*
–**Member** Society of Women in Physics *September 2023-present*
–**Member** Astronomy Club *September 2023-present*
–**Co-Manager** Sweets and More *January 2024-present*
–**Substitute Monitor/Teacher's Aide** Suffern Central School District *January 2025-present*
–**Assistant Manager** The Village Blend *September 2021-present*

Achievements

–**David Van Berkom Award** *Spring 2026*
–**International Scholars Program (ISP) Scholarship** *Spring 2026*
–**Lang Global Scholars Award** *Spring 2026*
–**Global Education Merit Scholarship** *Spring 2026*
–**NASA Spring Space Grant** *Spring 2026*
–**MA Space Grant Summer Fellowship** *Summer 2026*
–**MA Space Grant Summer Fellowship** *Summer 2024*
–**Chancellors Award** *2023-2027*